PIONEER JUNIOR COLLEGE, SINGAPORE

JC2 PRELIMINARY EXAMINATIONS
HIGHER 1

CANDIDATE NAME	

CT GROUP	**1**	**7**				INDEX NUMBER					

CHEMISTRY **8873/02**

Paper 2 Structured **11 September 2018**

2 hours

Candidates answer on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer only **one** question.

You are advised to show all working in calculations.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

FOR EXAMINER'S USE					
Paper 1		**Paper 2 Section A**		**Paper 2 Section B**	
Total	/ 30	4	/ 10	Total	/ 20
		5	/ 6		
Paper 2 Section A		6	/ 7	Penalty	s.f. / units
1	/ 4	7	/ 5	**TOTAL**	**/ 110**
2	/ 9	8	/ 8	**GRADE**	
3	/ 11	**TOTAL**	**/ 60**		

Answer **all** the questions in this section, in the spaces provided.

1 Fig. 1.1 shows a sketch of the logarithm of the first fourteen ionisation energies, log (IE), of element **A** against the number of electrons removed.

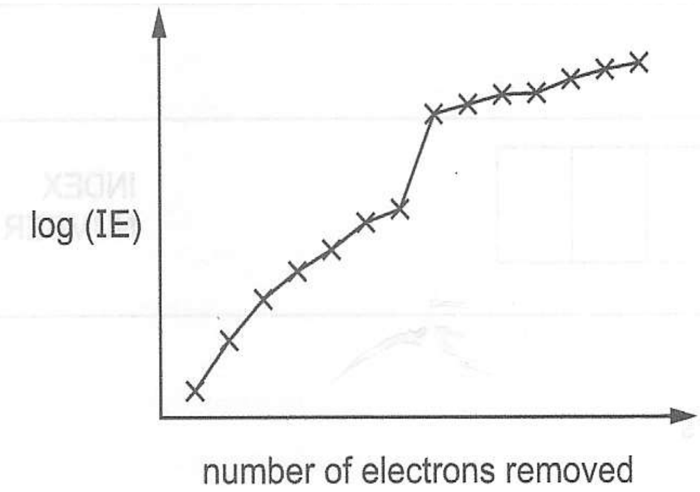

Fig. 1.1

(a) Write an equation for the second ionisation energy of element **A**. [1]

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(b) Explain the following features of Fig. 1.1.

(i) The increase in ionisation energies as the electrons are removed. [1]

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(ii) The significant jump in values from the 7th to the 8th ionisation energy. [1]

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(c) Sketch the shape of the orbital of element **A** that is not fully filled. [1]

[Total: 4]

2 When dry chlorine is passed over heated aluminium foil in a hard glass tube, a vapour is produced which condenses to a yellow-white solid in the cooler parts of the tube. At low temperatures the vapour has the empirical formula $AlCl_3$ and an M_r of 267.

(a) Suggest the molecular formula of the vapour, and draw a structure to describe its bonding. [2]

The yellow-white solid reacts with water in two separate ways.

- When a few drops of water are added to the solid, steamy white fumes are evolved and a white solid remains, which is insoluble in water.
- When a large amount of water is added to the solid, a clear, weakly acidic solution results

(b) Write equations, including state symbols, for these two reactions and explain the observations. [4]

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(c) Describe and explain what you would see when a sample of $Na_2O(s)$ and P_4O_{10} is added to a solution of litmus in water separately. Write equations for the reaction that occur. [3]

Note: The litmus solution is a purple solution when it is added to a neutral solution.

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[Total: 9]

3 **(a)** In a molecule of $SOCl_2$, the sulfur atom has four bonds.

Draw a 'dot and cross' diagram of $SOCl_2$. [1]

(b) When $SOCl_2$ is reacted with a carboxylic acid to produce an acyl chloride, two acidic gases are fomed.

$$SOCl_2(l) + RCOOH(l) \rightarrow ROCl(l) + SO_2(g) + HCl(g)$$

A 1.00g sample of a carboxylic acid RCOOH was treated in this way, and the gases were absorbed in 60.0cm^3 of 0.500 mol dm^{-3} NaOH(aq), in excess.

(i) Write equations for the reactions between NaOH and SO_2. [1]

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The excess NaOH was titrated with 0.500 mol dm^{-3} H^+(aq). It required 10.8 cm^3 of the H^+(aq) solution to reach the end-point.

(ii) Calculate the total amount, in moles of NaOH that reacted with the SO_2 and HC*l*. [2]

(iii) Calculate the amount, in moles of RCOOH that produced the SO_2 and HC*l*. [1]

(iv) Hence calculate the M_r of the carboxylic acid, RCOOH. [1]

(v) The R group contains carbon and hydrogen only. Suggest the molecular formula of RCOOH. [1]

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(c) The following synthetic route shows how a carboxylic acid can be converted into an amine.

$$CH_3CO_2H \xrightarrow{SOCl_2} CH_3COCl \xrightarrow{NH_3} CH_3CONH_2 \xrightarrow{\text{Step 3}} CH_3CH_2NH_2$$

(i) Suggest a reagent for step 3. [1]

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Angelic acid, $C_5H_8O_2$, is a natural product isolated from the roots of the angelica plant.

- Angelic acid reacts with H_2 + Ni to form **T**, $C_5H_{10}O_2$.
- **T** undergoes the above synthetic route to form the amine **U**, $C_5H_{13}N$.
- **U** can also be made by reacting 1-bromo-2-methylbutane with ammonia.

Angelic acid displays *cis-trans* isomerism.

(ii) Suggest the structures for angelic acid, **T** and **U**. [3]

Angelic acid	**T**	**U**

[Total: 11]

4 At 450K phosphorus(V) chloride, $PCl_5(g)$ decomposes to form phosphorus(III) chloride, $PCl_3(g)$, and chlorine, $Cl_2(g)$. A dynamic equilibrium is established as shown.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \qquad \Delta H = +124 \text{ kJ mol}^{-1}$$

(a) **(i)** With the use of a Boltzmann distribution curve, explain why an increase in temperature will increase the rate of decomposition of $PCl_5(g)$. [4]

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(ii) State and explain the effect of increasing temperature on the percentage of $PCl_5(g)$ that decomposes. [2]

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(b) Explain the meaning of the term dynamic equilibrium and the conditions necessary for it to become established. [2]

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(c) When 2.00 mol of PCl_5(g) are decomposed at 450K and 1.00×10^5 Pa in a 7.5 dm^3 container, the resulting equilibrium mixture contains 0.800 mol of Cl_2(g).

Calculate the value of K_c and state its units. [2]

[Total: 10]

5 **(a)** Magnesium(I) chloride, MgC*l*, is an unstable compound and readily decomposes as shown.

$$2MgCl(s) \longrightarrow Mg(s) + MgCl_2(s)$$

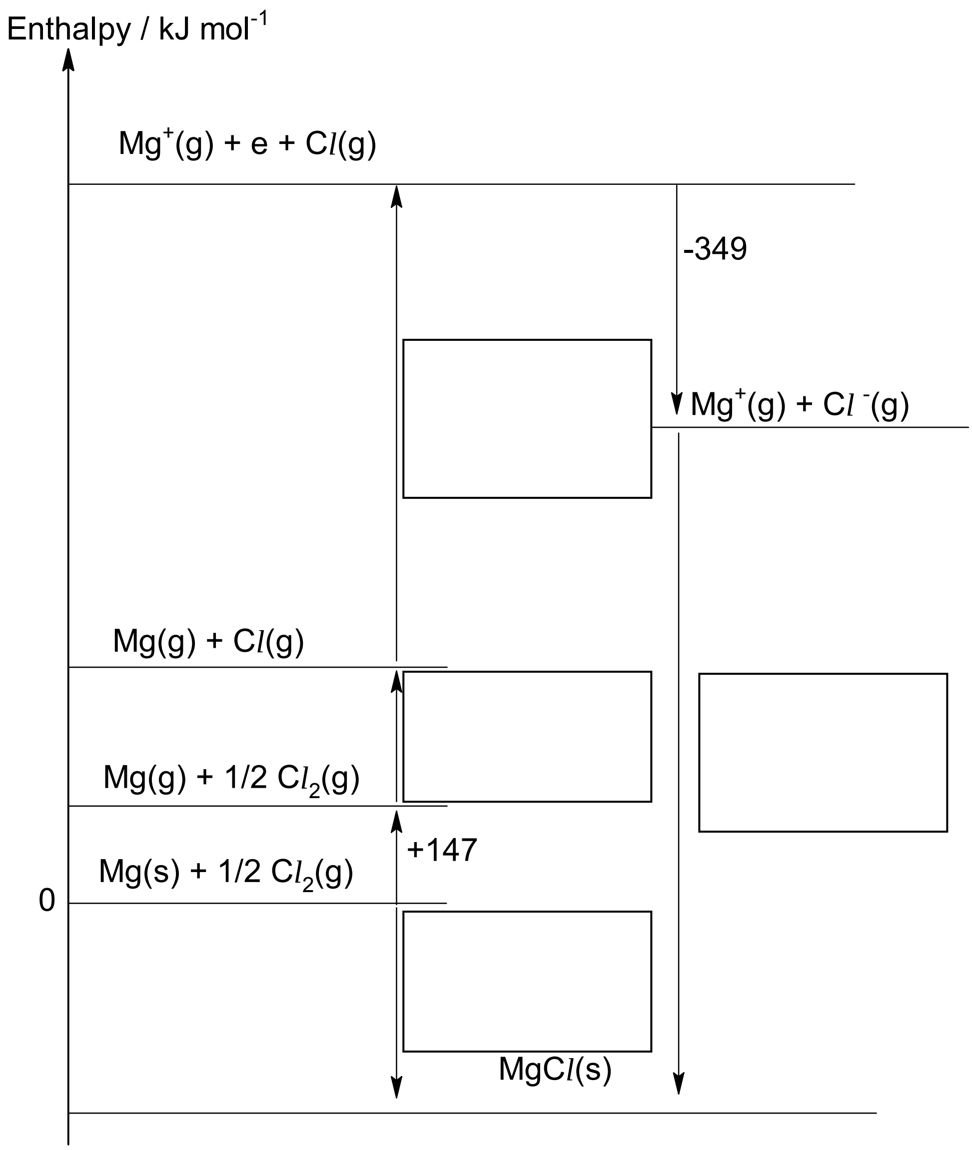
Use the following data to calculate the enthalpy change of this reaction. [1]

ΔH_f MgC*l*(s) = –106 kJ mol^{-1};
ΔH_f $MgCl_2$(s) = –642 kJ mol^{-1}

(b) **(i)** The equation for which ΔH is the lattice energy for MgC*l* is shown.

$$Mg^{+}(g) + Cl^{-}(g) \longrightarrow MgCl(s)$$

Label the relevant enthalpy changes in the energy cycle below. [2]



(ii) Use the energy cycle above and relevant data from the *Data Booklet* to calculate a value for the lattice energy of MgC*l*. [1]

(iii) Suggest how the lattice energy of $MgCl_2$ will compare to that of $MgCl$. Explain your answers. [2]

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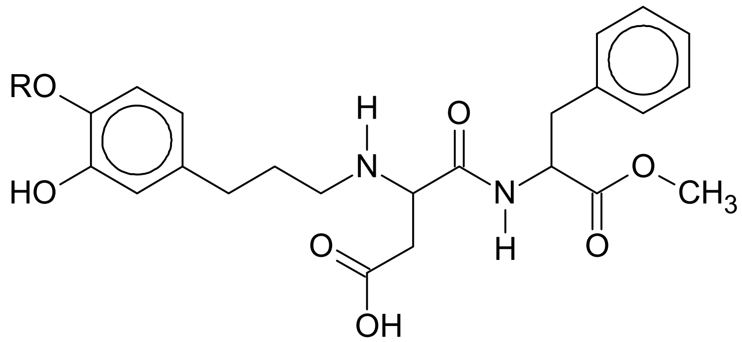
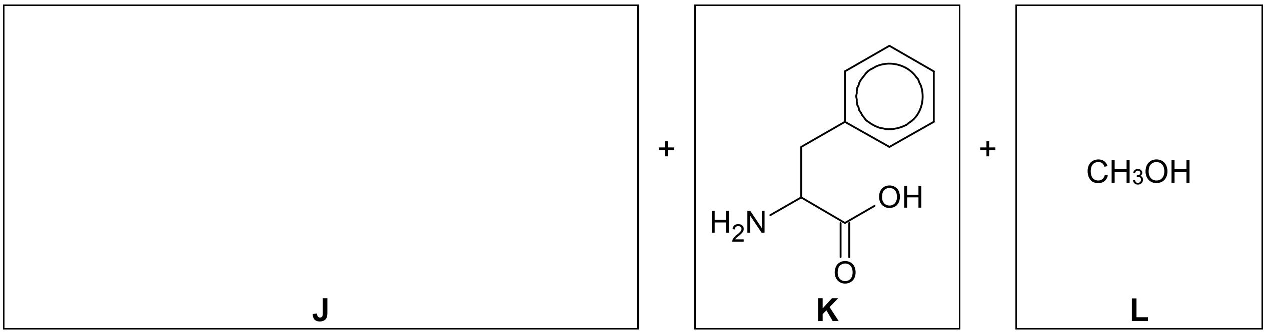
[Total: 6]

6 The compound *Advantame* is a sweetener that tastes approximately 25000 times sweeter than sucrose.



Advantame

(a) The decomposition of *Advantame* produces three molecules, **J**, **K** and **L**. The RO– group in *Advantame* is unreactive.



(i) Name the type of reaction occurring. [1]

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(ii) Draw the structure of **J** in the box above. [1]

(b) **(i)** State what you would observe when acidified potassium manganate(VII) is added to a solution of **L**. [2]

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(ii) Write balanced equation for the reaction taking place in **(b)(i)**. [1]

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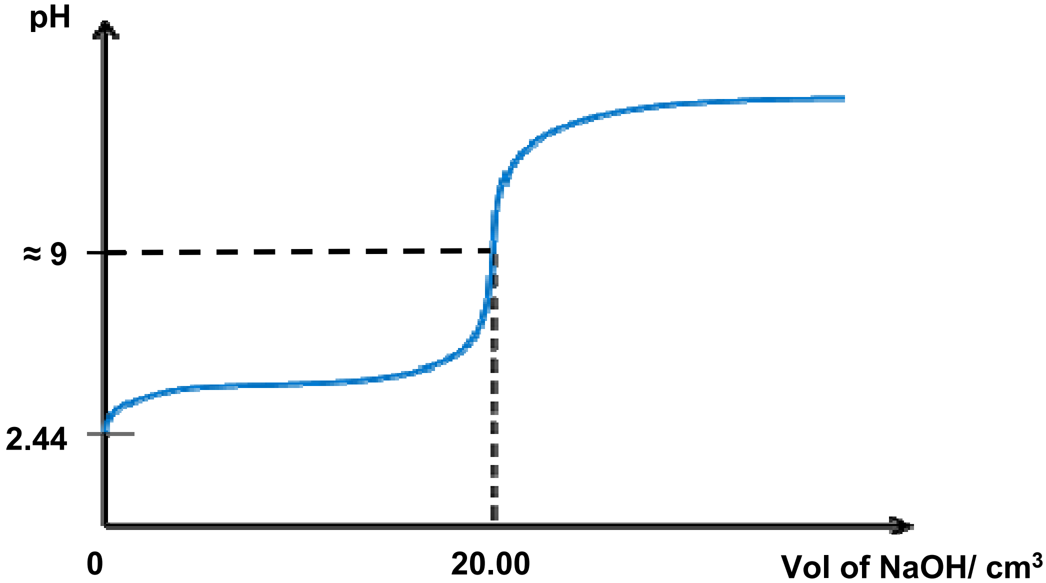
(c) **K** can be polymerised. Draw the structure of the polymer showing two repeat units. The linkage between the monomer units should be fully displayed. [2]

[Total: 7]

7 A 25.0 cm^3 of 0.72 mol dm^{-3} of CH_3COOH in a conical flask was titrated against aqueous sodium hydroxide and the pH of the solution in the conical flask was plotted against volume of NaOH(aq).



(a) **(i)** Using the graph, determine the concentration of aqueous sodium hydroxide in the burette. [1]

(ii) Using the data in the graph, show that ethanoic acid is a weak acid. [1]

(b) Circle the region on the graph on this question paper where a buffer solution is produced from the titration. [1]

(c) Phenolphthalein indicator is used in this titration. Using the information in the table provided below, give two reasons why phenolphthalein is a good indicator. [2]

Indicator	pH range	Colour change	
		Colour in acid < pH 8.2	Colour in alkali > 10.0
Phenolphthalein	8.2 – 10.0	Colourless	Red

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[Total: 5]

8 Many elements have *allotropes*. Carbon has a number of allotropes including diamond, graphite and graphene. Each of these allotropes has different properties. This means the allotropes can be used for different purposes.

Diamond is hard and does not conduct electricity.
Graphite is soft and has high electrical conductivity.
Graphene is a nanomaterial and has high electrical conductivity. A large layer of graphene would be very strong.

(a) **(i)** Describe what is meant by allotropes. [1]

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(ii) Describe the structure of graphene. Explain how this structure gives graphene the properties of high strength and high electrical conductivity.

You may find it useful to include a labelled diagram in your answer. [3]

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(b) Research in nanotechnology is growing and many new nanomaterials are being produced with ever more uses. Uses of nanoparticles and nanomaterials is relatively new and there are concerns about their direct impact on human health and environment.

For example, nanoparticles are used to deliver drugs within cells.

(i) State the difference between a nanoparticle and a nanomaterial in terms of size. [1]

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(ii) Suggest what properties of nanoparticles enable them to deliver drugs within cells. Explain your answer. [2]

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(iii) Give one example of an impact of nanoparticles on the environment. [1]

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[Total: 8]

Section B

Answer **one** question from this section, in the spaces provided.

1 If potassium iodide and ammonium peroxodisulfate are mixed in the presence of starch and sodium thiosulfate, the iodine liberated reacts with the thiosulfate ions until they are all used up.

$$S_2O_8^{2-} + 2I^- \rightarrow I_2 + 2SO_4^{2-}$$
$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

When no thiosulfate ions are left, any iodine formed reacts with starch and the dark blue colour of the starch-iodine complex appears. The time taken for the dark blue colour to appear with a constant quantity of sodium thiosulfate present depends on the rate of formation of iodine

Table of results

Experiment	**Volume of I^- / cm^3**	**Volume of $S_2O_8^{2-}$ / cm^3**	**Volume of deionised water / cm^3**	***Volume of starch / cm^3***	***t / s***
1	**10.00**	**20.00**	**0.0**	**1.0**	**23**
2	**10.00**	**10.00**	**10.0**	**1.0**	**46**
3	**10.00**	**10.00**	**10.0**	**1.0**	***40***

(a) **(i)** Using the results in your table, deduce the order of the reaction with respect to the peroxodisulfate ions. Explain clearly how you arrived at your answer. [2]

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(ii) Instead of washing the beaker as required before performing experiment **3**, a student simply just poured away the reaction mixture. There were some left over reaction mixture in the beaker when he performed experiment **3**. State and explain why the time is shorter than in experiment **2**, assuming all other conditions are kept constant. [1]

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(iii) Explain why a fixed amount of sodium thiosulfate is required. [1]

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(b) It was reported that the order of reaction with respect to iodide ions is one.

In order to substantiate this information, the chemistry teacher gives out these instructions to his class.

1) Prepare a solution mixture containing 0.100 mol dm^{-3} potassium iodide and 1.00 $moldm^{-3}$ of ammonium peroxodisulfate.

2) Record the concentration of the iodide ions against time from the start of the reaction.

3) Plot a graph of concentration of iodide ions against time

(i) Sketch the graph of concentration of iodide ions against time, given that the half-life is 150 seconds, showing the values and important features of your graph to show that it is first order reaction with respect to iodide. [2]

(ii) A student mistakenly prepared a solution containing 1.00 mol dm^{-3} potassium iodide and 0.100 mol dm^{-3} of ammonium peroxodisulfate instead.

He recorded the concentration of the peroxodisulfate ions against time from the start of the reaction and plot a graph of concentration of peroxodisulfate ions against time on a piece of graph paper.

On realising the mistake, the teacher instructed him to carry another experiment by preparing a solution containing 2.00 mol dm^{-3} potassium iodide and 0.100 mol dm^{-3} of ammonium peroxodisulfate.

He recorded the concentration of the peroxodisulfate ions against time and plot the data on the same graph paper.

Based on the data from both experiments on the graph paper, describe how the student can determine that it is first order reaction with respect to iodide ions. [2]

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(c) It is reported that Fe^{3+} catalyses the reaction between ammonium peroxodisulfate and iodide ions.

Explain how Fe^{3+} acts as a catalyst. [2]

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(d) The table below gives the melting points, in ^{o}C, of the fluorides and chlorides of two elements in Period 3.

	iron	silicon
chlorides	677	-70
iodides	587	120.5

Explain, in terms of structure and bonding, the differences in melting point between

(i) $FeCl_2$ and $SiCl_4$ [2]

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(ii) SiI_4 and $SiCl_4$ [2]

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(e) The Harcourt and Esson reaction is a redox reaction just like the reaction between I^- and $S_2O_8^{2-}$. The type of reaction also occurs in organic, as the following transformations from

$$\underset{\text{propan-1-ol}}{CH_3CH_2CH_2OH} \longleftrightarrow \underset{\text{propanal}}{CH_3CH_2CHO} \longrightarrow \underset{\text{propanoic acid}}{CH_3CH_2CO_2H}$$

(i) State a reagent you could use to convert propanoic acid into propan1-ol. [1]

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(ii) Both propanal and propanoic acid can be obtained from propan-1-ol.

State the reagents you could use to carry these reactions. [2]

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(iii) During the reduction of propanoic acid to propan-1-ol, it was suspected that there are traces of unreacted propanoic acid together with propan-1-ol after the reaction.

Suggest a chemical test to confirm the presence of the unreacted propanoic acid, state the observation that you will see. [2]

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(iv) State the type of reaction that has occurred in **(e)(iii).** [1]

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[Total: 20]

2 The halogens are a group in the Periodic Table consisting of five chemically related elements: fluorine, chlorine, bromine, iodine and astatine. The group of halogens is the only periodic table group that contains elements in all three main states of matter at standard temperature and pressure.

(a) Explain the trend in boiling points of halogens down the group. [2]

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(b) Ethene can react with halogens like chlorine and bromine to form 1,2–dichloroethane and 1,2–dibromoethane respectively.

(i) Describe the bonding between the two carbon atoms in ethene in terms of orbital overlap. You may draw a diagram to illustrate your answer. [2]

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(ii) The enthalpy change of combustion of ethene is given as –1390 kJ mol^{-1}. Calculate the mass of water at 30°C that can be brought to the boiling point by burning 2.0 dm^3 of ethene.

Assume that 75% of the heat from the ethene is absorbed by the water and volume of gas is measured at room temperature and pressure. [2]

(iii) When burnt in a limited supply of air, ethene forms carbon dioxide, carbon monoxide and water according to the equation below.

$$C_2H_4(g) + 5/2O_2(g) \longrightarrow CO_2(g) + CO(g) + 2H_2O(l)$$

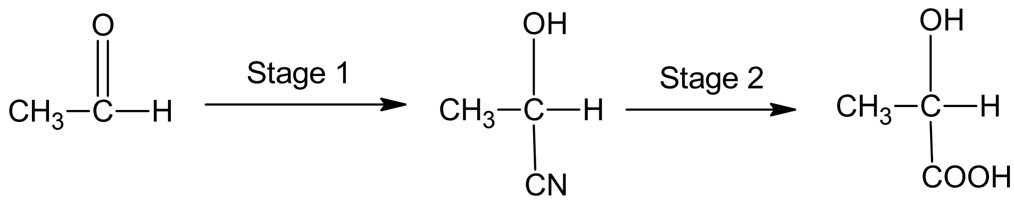
$$2CO_2(g) + 2H_2O(l)$$

Calculate the ΔH of the reaction given the energy cycle and following data.

ΔH_c (ethene) = –1390 kJ mol^{-1};
ΔH_f (carbon dioxide) = –394 kJ mol^{-1}
ΔH_c (carbon monoxide) = –283 kJ mol^{-1} [1]

(iv) Ethene can be converted to ethanal. State the reagents, conditions needed and the intermediate products formed in the conversion. [3]

(c) Ethanal undergoes a 3 stage synthesis to form lactic acid as follows.



(i) Name the type of reaction in Stage 1. [1]

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Lactic acid, $CH_3CH(OH)COOH$ is a monoprotic acid.

(ii) Give the equation which represent the dissociation of lactic acid in water. Identify the acid conjugate base pairs in the equation. [2]

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(iii) Explain, using equations, why an aqueous mixture of lactic acid and sodium lactate can act as a buffer solution

(I) on the addition of acid,

(II) on the addition of alkali. [2]

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(e) Lactic acid, $CH_3CH(OH)COOH$ is the raw material for the polymer, "polylactide" or PLA.

(i) What type of reaction takes place in this polymerisation? [1]

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One of the reasons PLA has attracted so much attention is that it is biodegradeable. This does, however, restrict some potential uses. The simple polymer has a melting poing of around 175°C, but softens between 60-80°C. However, its thermoplastic properties enable it to have a range of uses in fibres and in food packaging.

(ii) Explain why PLA would not be a suitable packing material for foods pickled in vinegar. [1]

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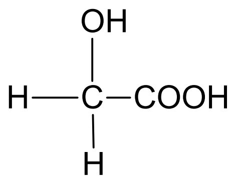
Lactic acid can also be co-polymerised with glycolic acid.



glycolic acid

(iii) Draw a section of the co-polymer showing one repeat unit. [1]

(iv) Suggest what type of bonding will occur between chains of this co-polymer, indicating the groups involved. [1]

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[Total: 20]

End of Paper